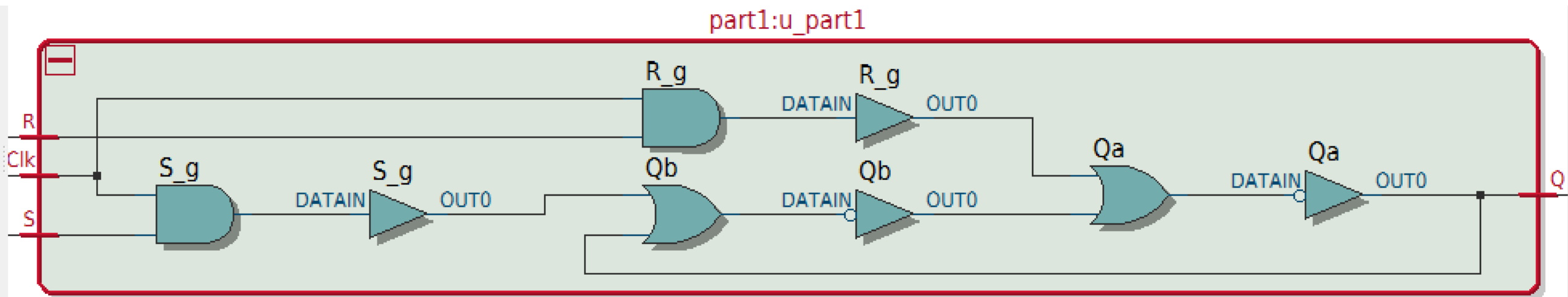
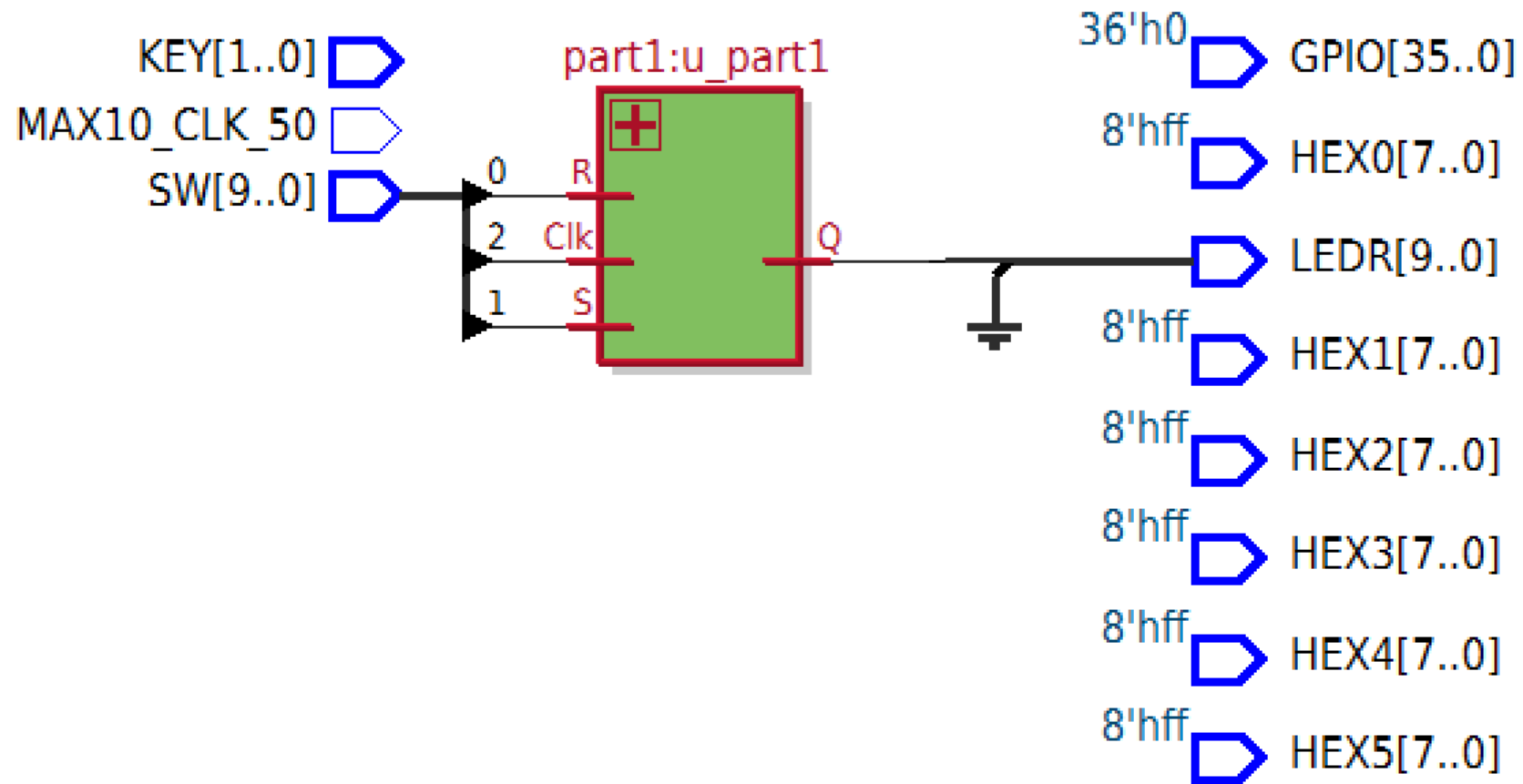


Part I: RTL Viewer of Gated RS Latch

- Inputs & Outputs: R (Reset), S (Set), Clk (Clock), and Q (Output).
- Gating Logic: Two AND gates that generate the gated signals R_g and S_g when Clk is high.
- Cross-Coupled Latch Core: OR gates combined with inverted buffer stages implementing the cross-coupled NOR logic to form Qa and Qb.

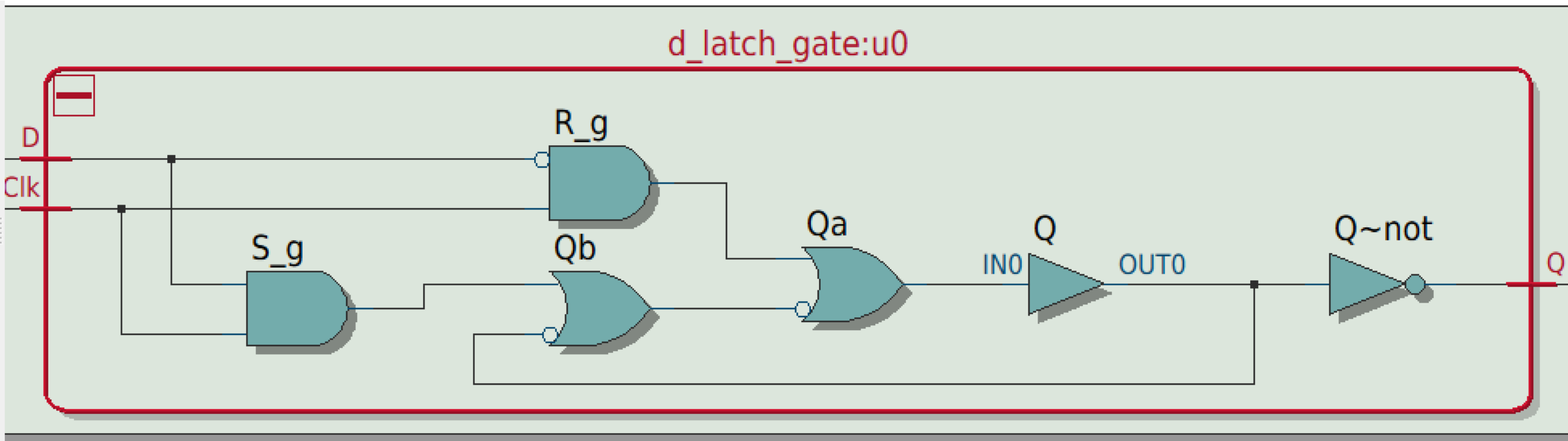


Part I: Top-Level RTL Viewer Diagram



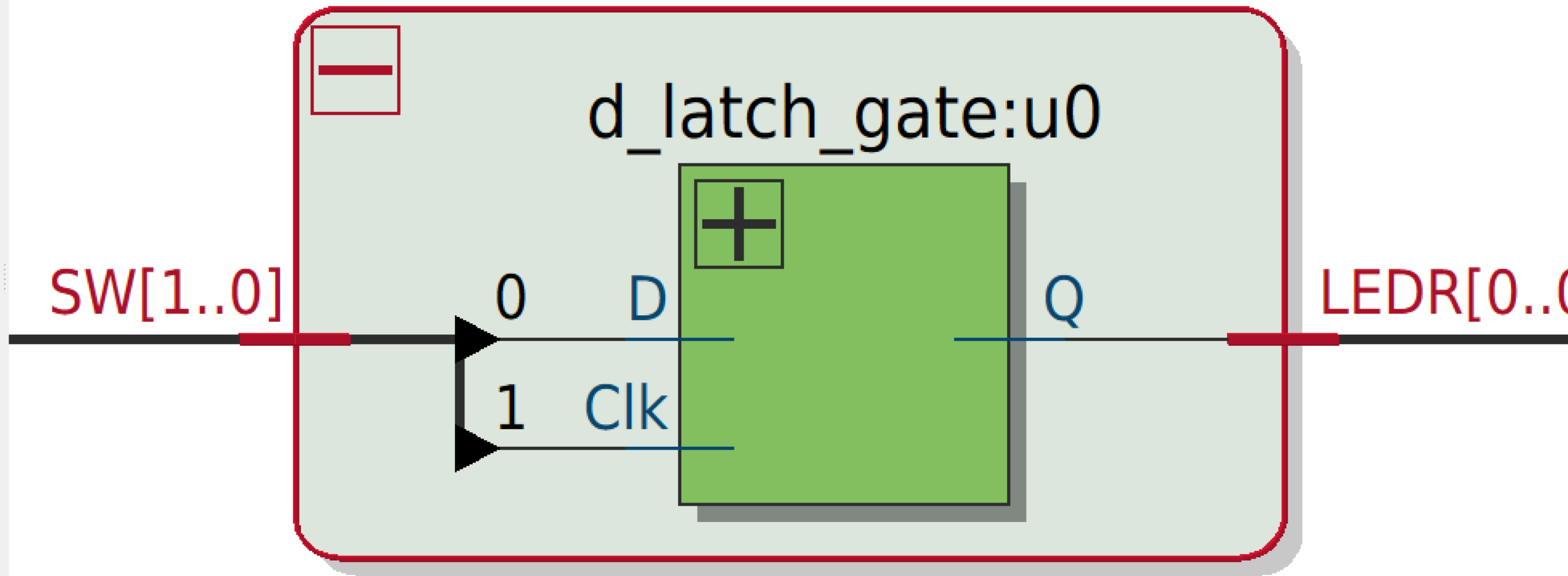
Part II: RTL Viewer D-Latch Gate

- **Submodule & Ports:** The expanded d_latch_gate instance u0 taking inputs D and Clk, and driving output Q.
- **Gating Logic:**
 - S_g: AND gate combining D and Clk to generate the gated set signal $S_g = D \times \text{Clk}$.
 - R_g: AND gate with an inverted D input combining inverted D and Clk to generate the gated reset signal $R_g = \text{inverted } D \times \text{Clk}$.
- **Latch Storage Core:** The cross-coupled NOR logic stages (Qb and Qa) that form the memory element.

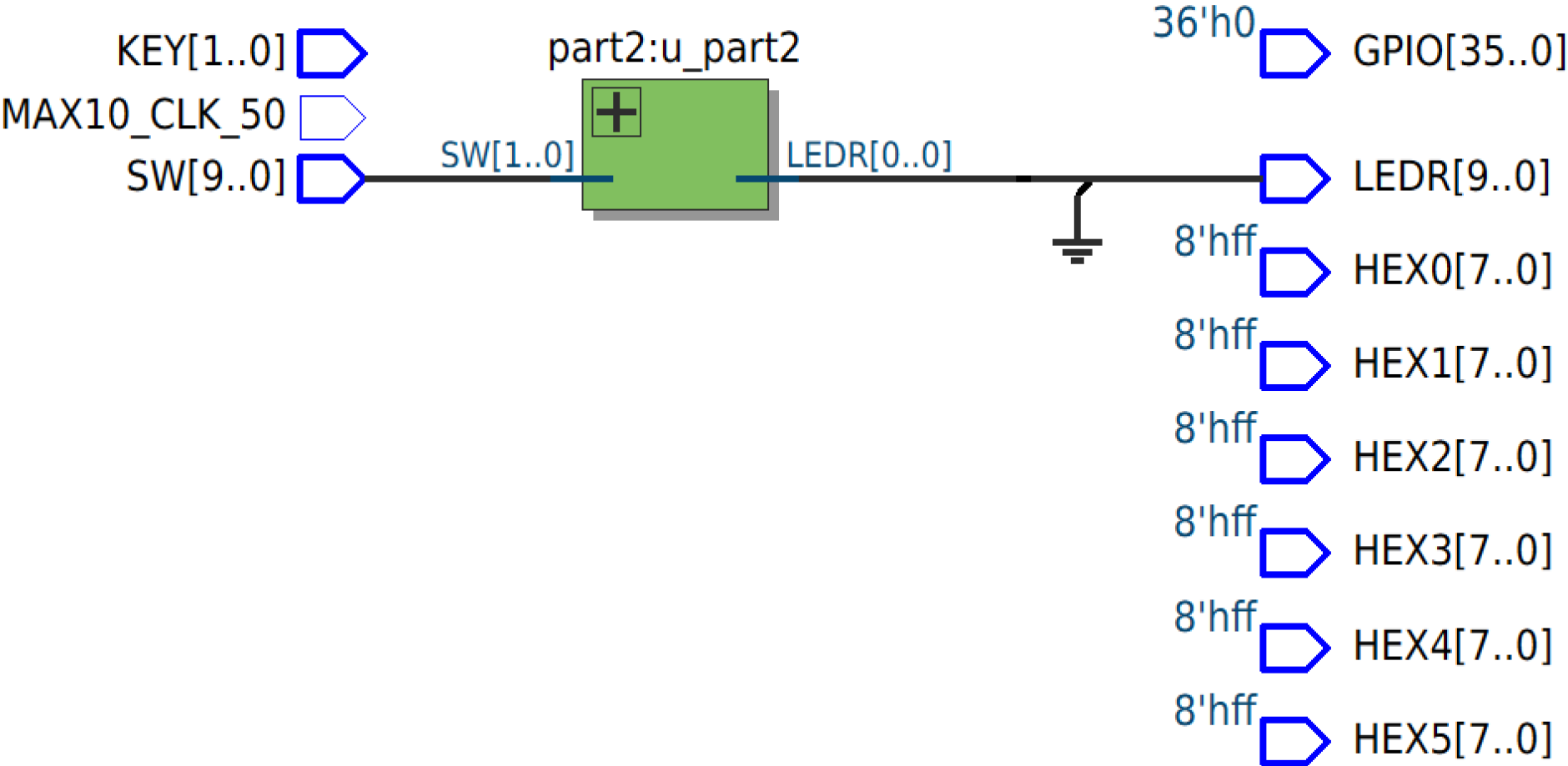


Part II: RTL Viewer Block Diagram

part2:u_part2

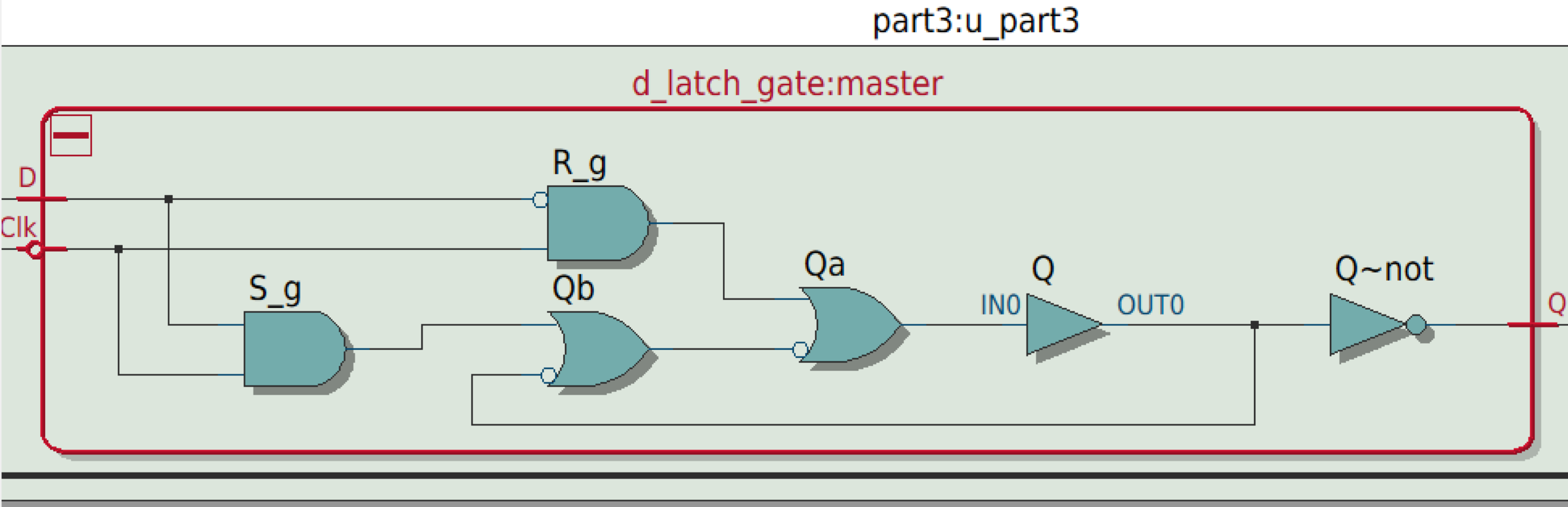


Part II: Top-Level RTL Viewer Diagram



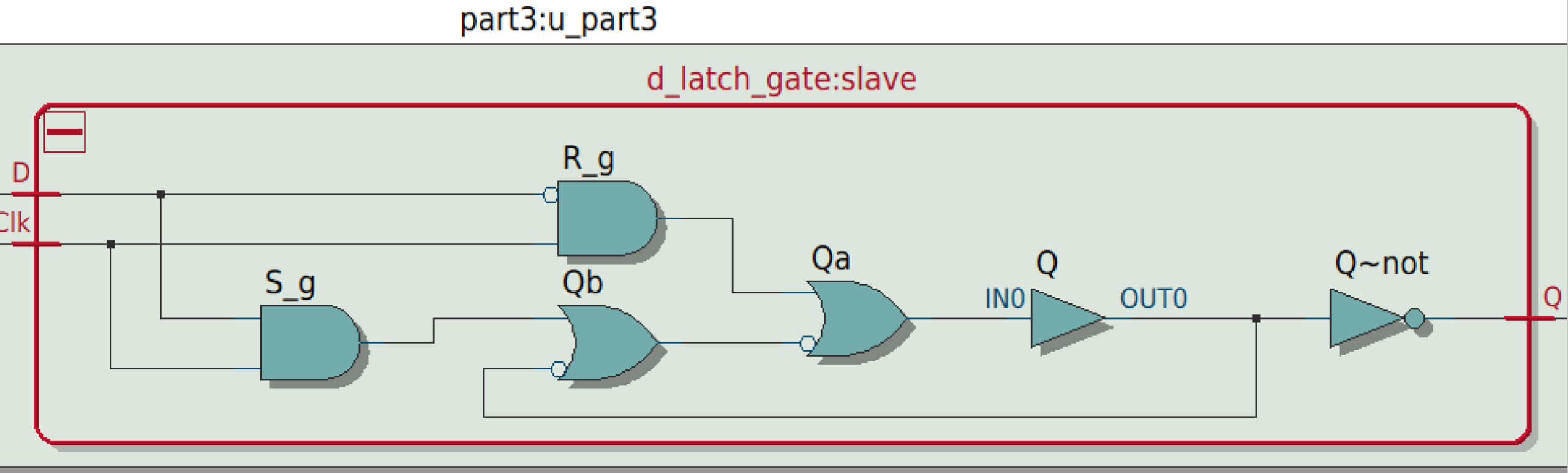
Part III: RTL Viewer Master D-Latch

- **Inverted Clock Input (Clk):** Notice the small bubble/circle on the Clk input port on the left boundary. This indicates an inverted clock input inverted Clk, meaning the master latch is active when the main Clock signal is **LOW**.
- **Gate-Level Logic:** Uses the same gated D-latch logic as Part II (AND gates for S_g and R_g, cross-coupled NOR logic for Qa and Qb).

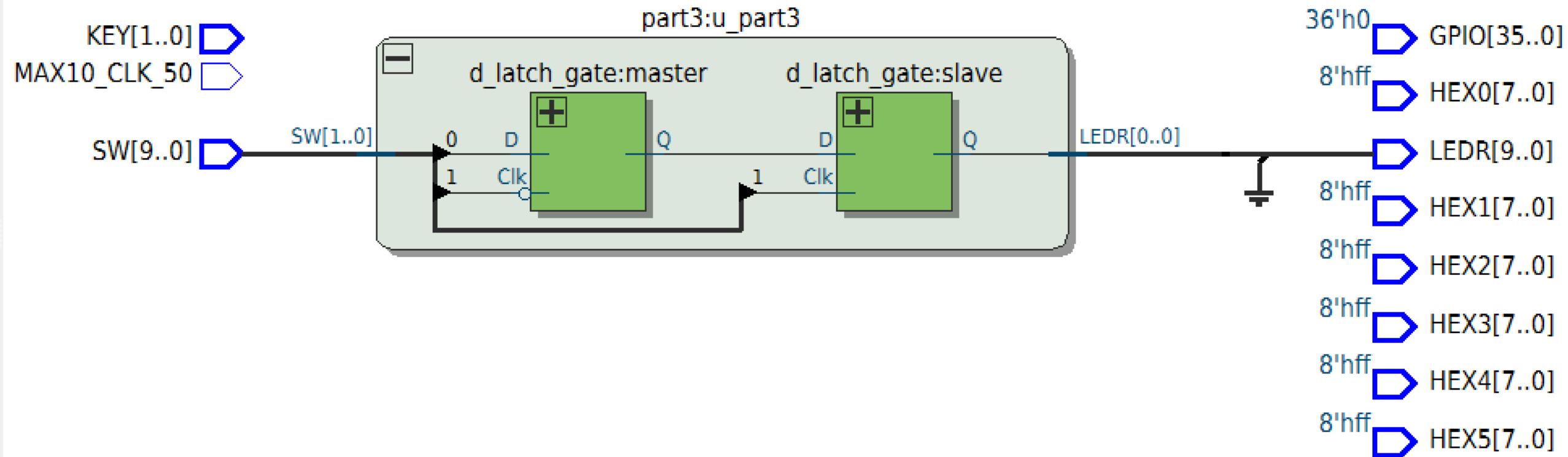


Part III: RTL Viewer Slave D-Latch

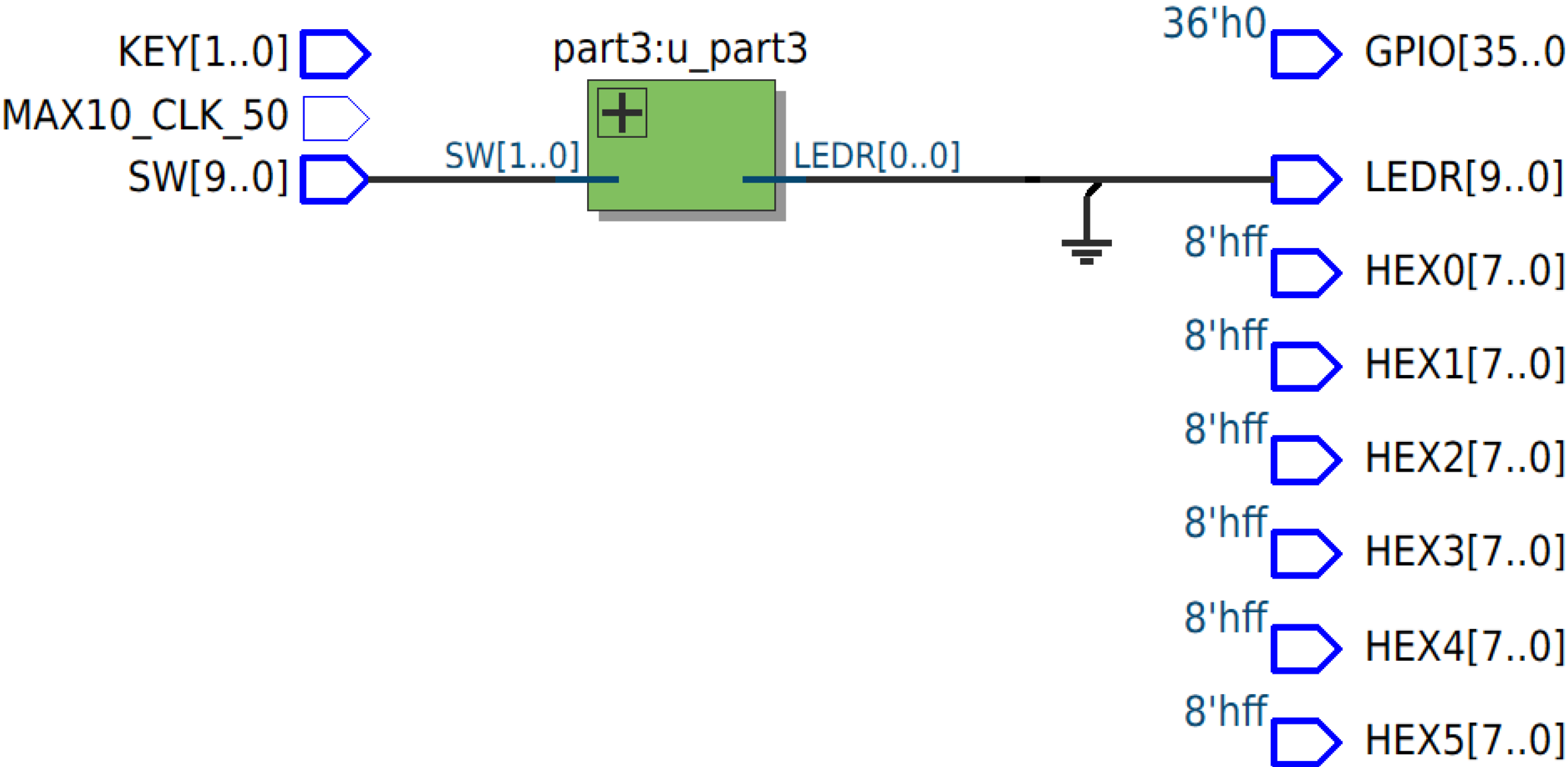
- **Non-Inverted Clock Input (Clk):** Notice that there is no inversion bubble on the Clk input port on the left boundary. This means the slave latch is active when the main Clock signal is **HIGH**.
- **Cascaded Data Input (D):** The data input D on this slave block is fed directly from the output of the master latch stage (Q_m).



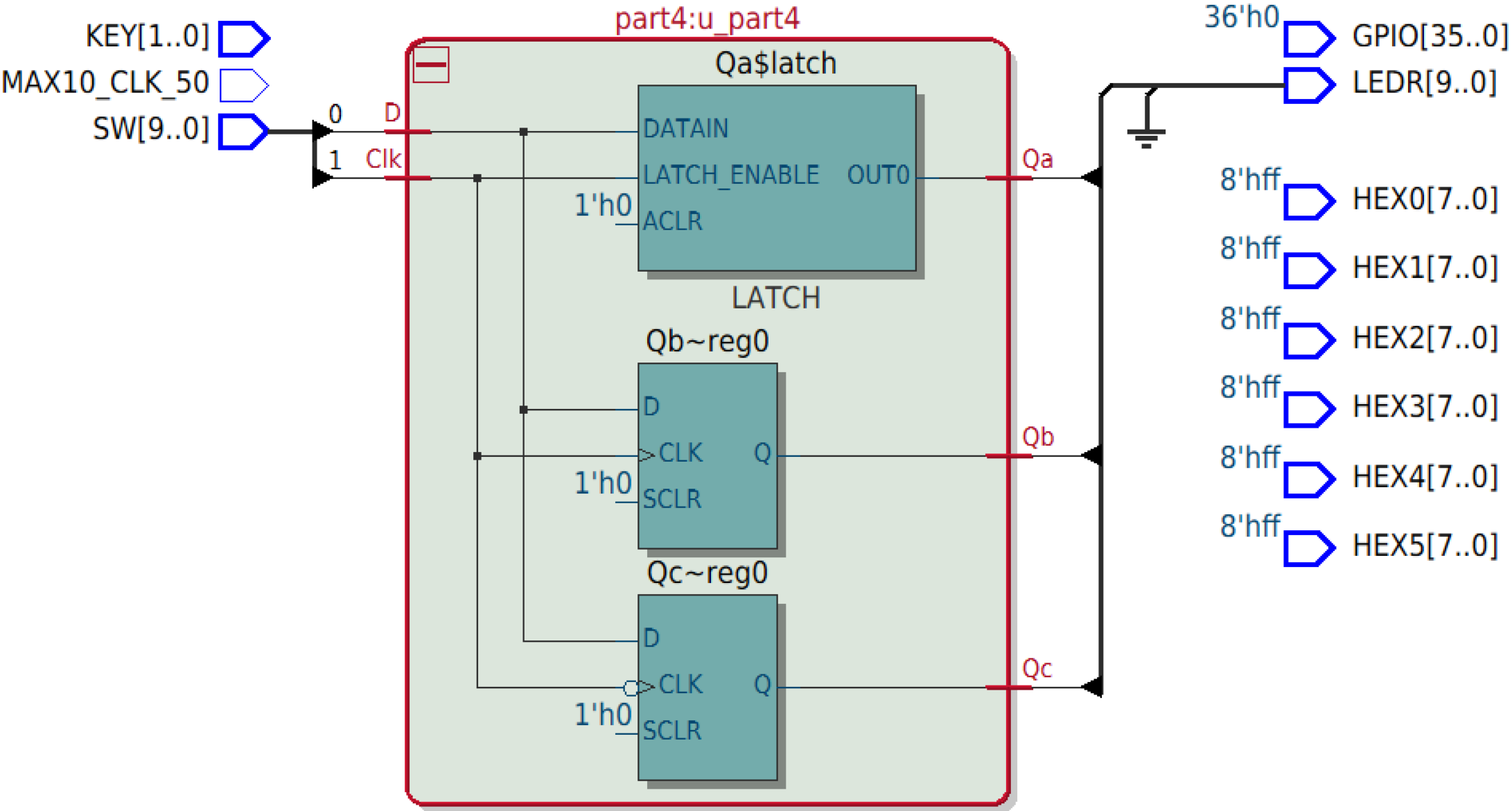
Part III: RTL Viewer Block Diagram



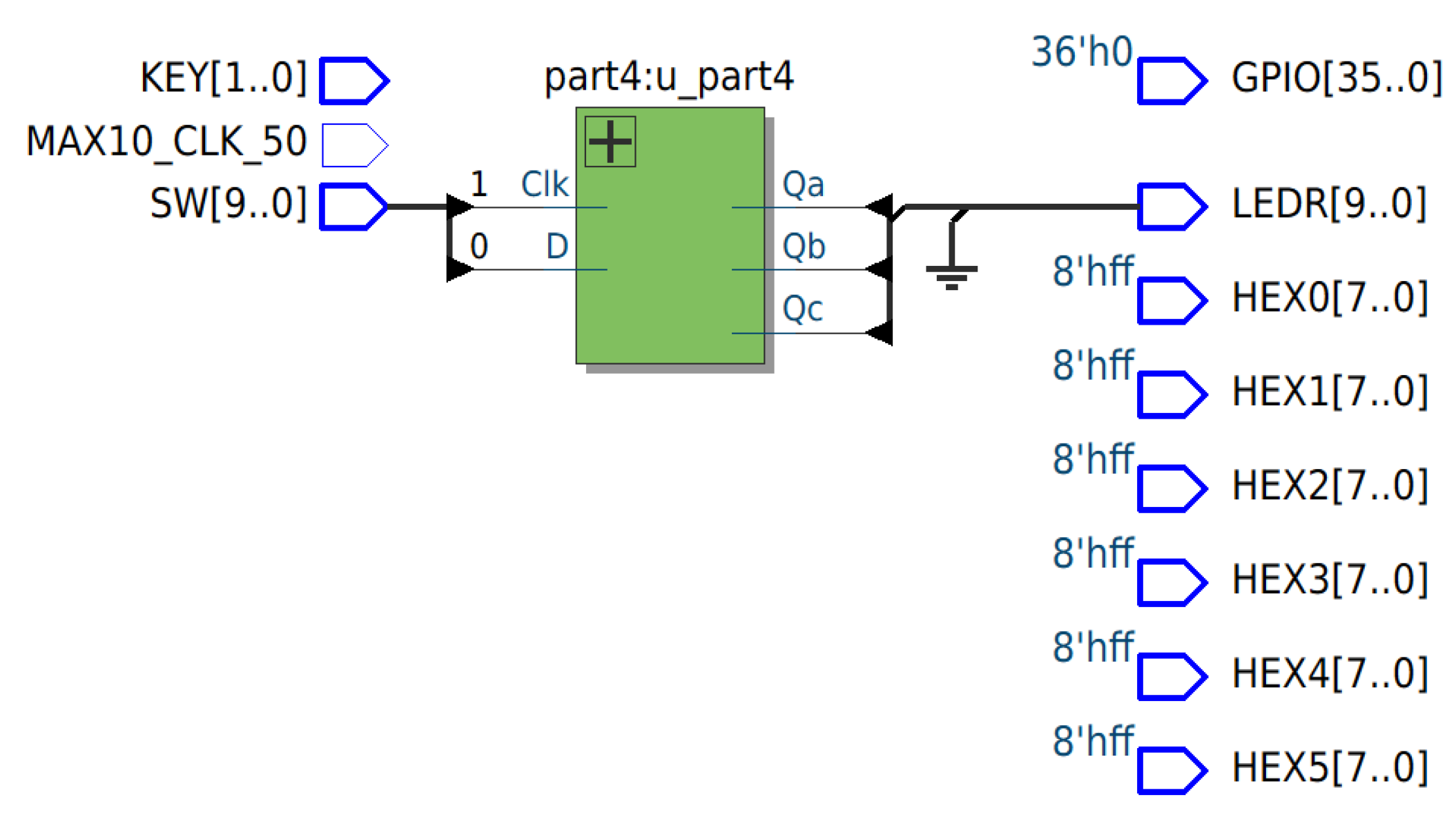
Part III: Top-Level RTL Viewer Diagram



Part IV: RTL Viewer Block Diagram

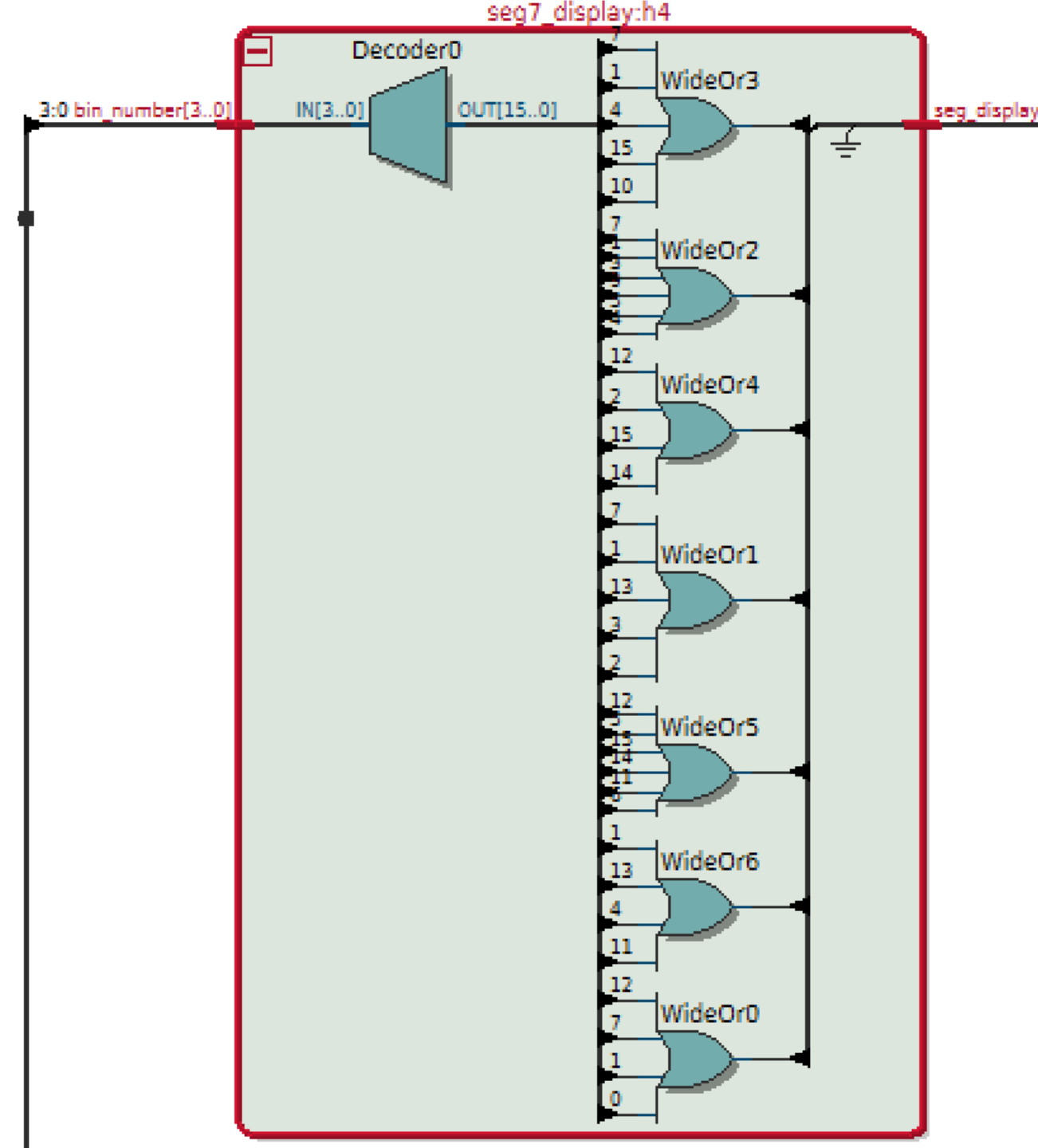


Part IV: Top-Level RTL Viewer Diagram



Part V: RTL Viewer 7 Segment Display

- **Binary Decoder (Decoder0):** Takes the 4-bit `bin_number[3..0]` input and decodes it into a 1-of-16 representation (`OUT[15..0]`) corresponding to hex digits 0 through F.
- **OR Gate Array (WideOr0 – WideOr6):** Multi-input OR gates that group the decoded hex lines to control each individual segment (A through G) based on your case statement.
- **Output Bus (seg_display[7..0]):** The 8-bit output signal driving the physical 7-segment LED pins, with bit 7 tied to ground (GND)



Part V: Top-Level RTL Viewer Diagram

